RETAIL SALES & INVENTORY INTELLIGENCE PROJECT REPORT

Using SQL and POWER BI

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Introduction

Retail businesses generate huge amounts of transactional and customer data every day. Without proper analytics, businesses cannot identify profitable products, customer behavior, sales trends, or inventory performance. This internship project focuses on building an interactive Retail Sales & Inventory Intelligence Dashboard using SQL and Power BI. The dashboard transforms raw business data into meaningful insights and helps management make data-driven decisions.

Business objectives

The primary objective of this project is to help the retail company improve sales performance, inventory management, customer understanding, and staff productivity by creating an interactive dashboard, using data analytics and business intelligence tools.

The company can use this dashboard to:

- ✓ Improve sales decisions
- ✓ Identify top-selling products
- ✓ Monitor inventory levels
- ✓ Understand customer behaviour
- ✓ Track staff performance
- ✓ Reduce shipment delays
- ✓ Improve store performance

Business problems

The organization faced several analytical and operational problems:

1. Difficulty in identifying top-selling products.
2. Poor inventory visibility across stores.
3. Some stores faced stock shoratges.
4. Customer buying behavior was not properly tracked.
5. No centralised reporting system.
6. Time-consuming manual analysis.
7. Orders were delayed.

Project workflow

The project workflow followed these stages:

CSV Files → Excel Cleaning → SQL Import → Create ER Relationships → Write SQL Queries → Create SQL Views → Power BI Connection → Build Dashboard → Business Insights → Recommendations

Dataset information

The project dataset consisted of multiple tables:

Table	Purpose
orders	order information
order_items	products inside orders
customers	customer details
staffs	employee details
stores	store information
products	product information
brands	brand details
categories	product categories
stocks	inventory quantity

Excel Data Cleaning Process

Before importing data into SQL, the CSV files were cleaned using Excel.

Step I: Checking null values and blank rows

- In discount, replaced null with 0

Step II: Removed duplicate records

- Using, **REMOVE DUPLICATES** in Excel

Step III: Standarized product and category names

Step IV: Remove extra spaces

- Like, “Santa Cruz “ to “Santa Cruz” using, **TRIM function()** in Excel.

Step V: Data format correction

- order_date & shipped_date were converted to Date format.

Step VI: Checking data types

Column Type	Example
Numeric	quantity, sales
Date	order_date
Text	product_name

Step VII: Removed unnecessary columns

Step VIII: Saved cleaned files

- in this name format, for eg., orders_clean.csv
  products_clean.csv

Cleaning was done before importing data into SQL to-

- ✓ improve dashboard accuracy
- ✓ reduce data errors
- ✓ improve SQL relationships
- ✓ create correct KPIs

SQL Database Creation

After cleaning files in excel, a database in SQL was created.

Step I: Database creation

```
CREATE DATABASE retail_project;
USE retail_project;
```

Step II: Imported CSV files

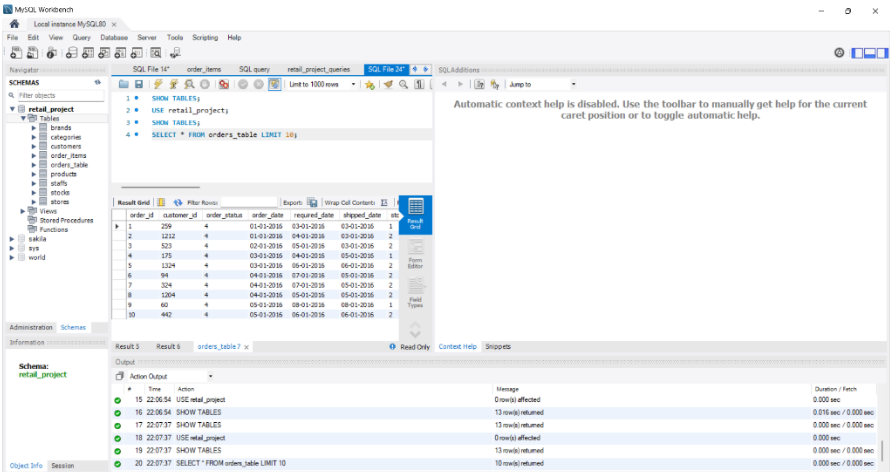
- Using Table Data Import Wizard

Step III: Verify Tables

- Using SQL query `SHOW TABLES;`

Step IV: Check Data Tables

```
SELECT * FROM orders_table LIMIT 10;
```



Step V: Establish ER relationships

Relationships were established between tables using primary keys and foreign keys.

Major Relationships:

• customers → orders (as one customer can place many orders)

• orders → order_items (One order can contain many products)

• products → order_items (One product can appear in many orders)

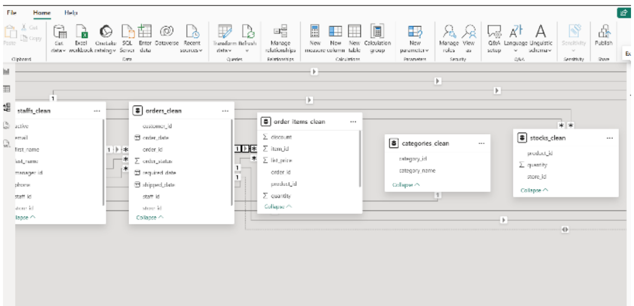
• categories → products (One category has many products)

• brands → products (One brand has many products)

• stores + products → stocks (Inventory tracking)

• stores → orders (One store handles many orders)

• staffs → orders (One staff handles many orders)



Step VI: SQL analysis and views

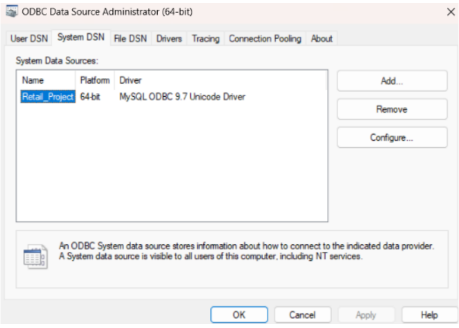
Basic SQL queries were used for:

- sales analysis
- inventory tracking
- customer analysis

- order analysis
  examples- Total sales, total products & sales by products.

Step VII: SQL connected to POWER BI

- using, ODBC Connector



POWER BI Dashboard Creation

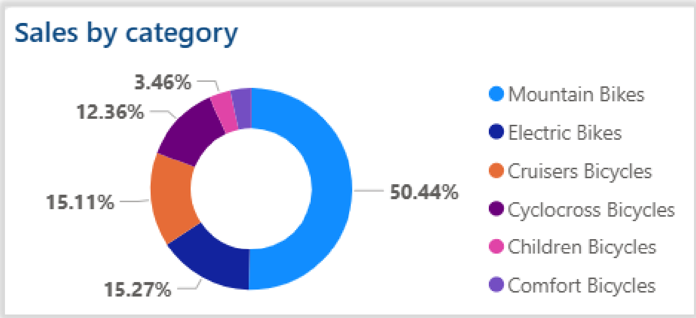
The dashboards included KPI cards, charts, slicers, filters, and data visualizations.

Several DAX measures were used, for eg.

TOTAL SALES Total Sales = SUM(order_items_clean[sales_amount])

DASHBOARDS CREATED

1. **Overview**
   - **Provides quick business insights and company overview.**
2. **Sales Analysis**
   - **Helps identify high-performing products, stores, and categories.**
3. **Inventory Analysis**
   - **Helps monitor inventory levels and stock shortages.**
4. **Customer Analysis**
   - **Helps understand customer demographics and buying behavior.**



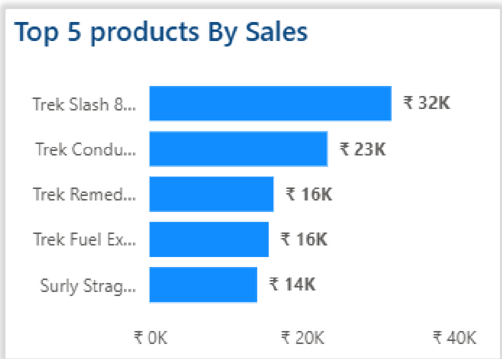
Which categories generate the most revenue?

Insights:

Half of the total sales come from one category. Mountain Bikes are the most demanded products. Mountain Bikes dominate customer demand and are the primary revenue driver.

Business decision possible:

1. Increase marketing for low-selling categories
2. Keep high stock for Mountain Bikes.



Highest revenue-generating products?

Insights:

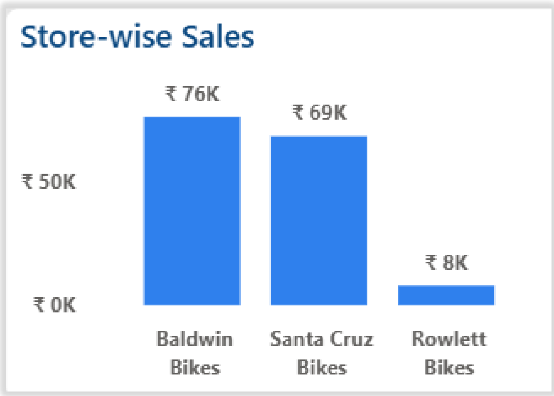
The Trek brand dominates sales performance. Premium bike products generate the highest revenue. Customers prefer high-quality bikes.

Top Products:

- Trek Slash 8
- Trek Conduit+
- Trek Remedy

Business decision possible:

1. Increase stock for Trek products.
2. Launch accessories bundle with top bikes.



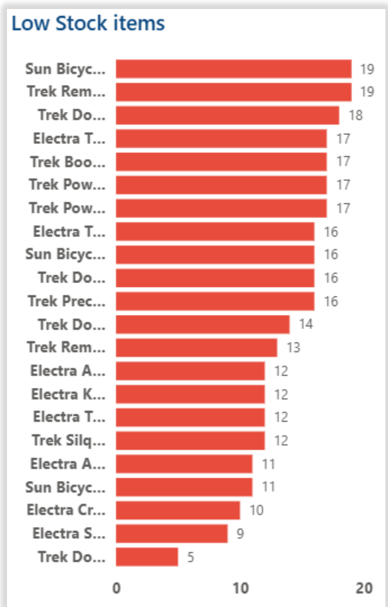
<u>Which store performs best?</u>

Insights:

Baldwin Bikes is the strongest revenue-generating store (₹76K). Rowlett bikes are underperforming.

Business decision possible:

1. Improve marketing in Rowlett.



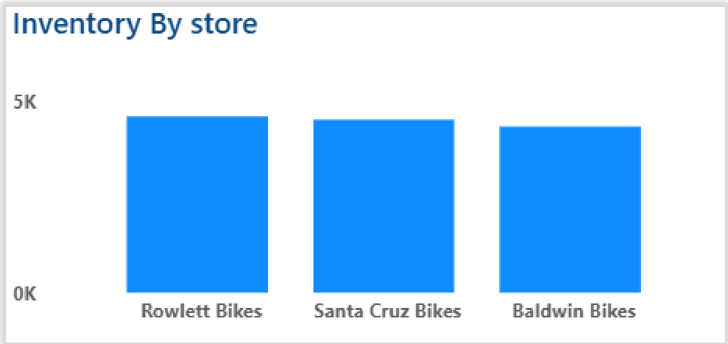
<u>Products needing immediate restocking?</u>

Insights:

High-demand products are running out quickly. Stock-out risk is very high for premium products. The risk of stock-outs can reduce customer satisfaction and sales.

Business decision possible:

1. Set automatic low-stock alerts
2. Increase safety stock levels for fast-moving items



Which store holds the highest stock?

Insights:

Inventory distribution across stores is fairly balanced. Baldwin generates strong sales with comparatively optimised inventory. Rowlett has high stock despite lower sales performance. Inventory allocation is focused mainly on high-demand categories.

Business decisions possible:

1. Shift excess stock from Rowlett to Baldwin.
2. Optimise inventory based on store demand.
3. Prevent unnecessary stock accumulation.

Customers who placed the highest number of orders?

Insights:

Highly active customers were-

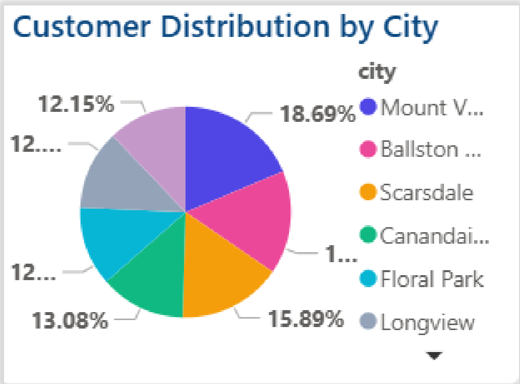
- Aleta

- Genoveva
- Jamaal
- Latasha
- Lorrie

Each customer placed 5 orders, which is higher than that of other customers and shows customer loyalty.

Business decision possible:

1. Launch loyalty reward programs.
2. Provide personalised offers.
3. Maintain strong customer relationships.



Which cities contribute the highest number of customers?

Insights:

Mount Vernon had the highest customer share. The business has geographically diversified customers, reducing dependency on one market.

Business decisions possible:

1. Launch local promotions
2. Expand operations in strong markets

BUSINESS RECOMMENDATIONS

1. **Improve inventory management**
   - Rebalance stock over stores.
   - Set automatic low-stock alerts.

     Benefit-
     - ✓ Lower shortage cost
     - ✓ Better stock utilization
2. **Focus on High-demand products**

- Mountain bike inventory.
- Premium bike promotions.

  Benefit-

  - ✓ Better profit margins.
  - ✓ Higher sales revenue

3. **Strengthen customer retention**
   - Introduce loyalty programs.
   - Membership discounts.

     Benefit-

     - ✓ Repeat purchases
     - ✓ Higher customer lifetime value

4. **Improve demand forecasting**
   - Analyse sales trend and seasonal demand, before inventory planning.

     Benefit-

     - ✓ Reduce overstocking
     - ✓ Improve sales forecasting

CONCLUSION

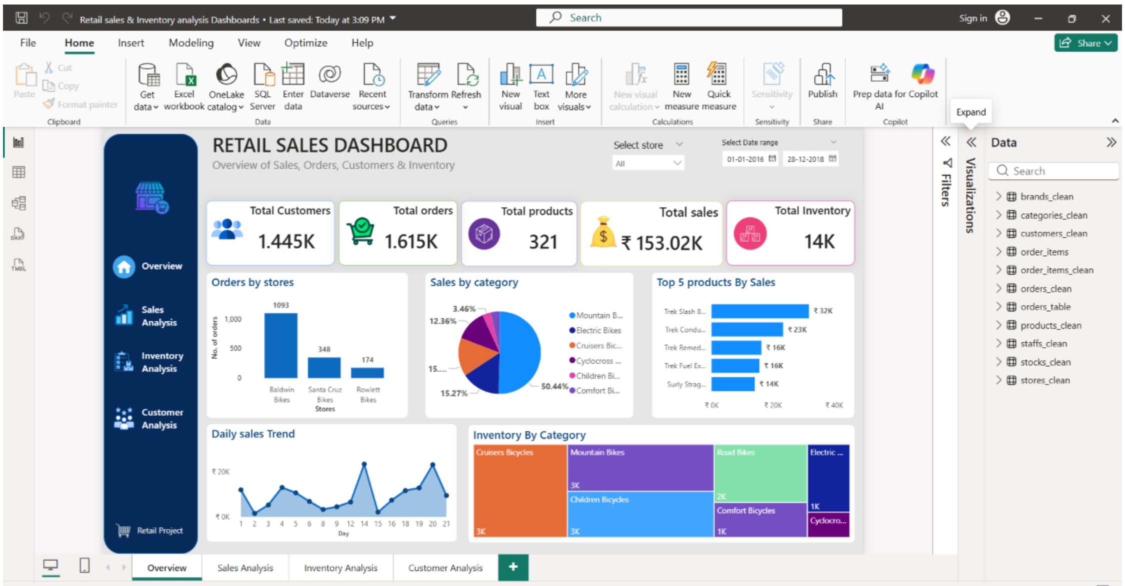
This project helped the business understand its overall performance in a much clearer and smarter way through data analysis and interactive dashboards. By analysing sales, inventory, customer behavior, products, and store performance, the company was able to identify which stores and products perform best, which products require restocking, and where operational improvements are needed.

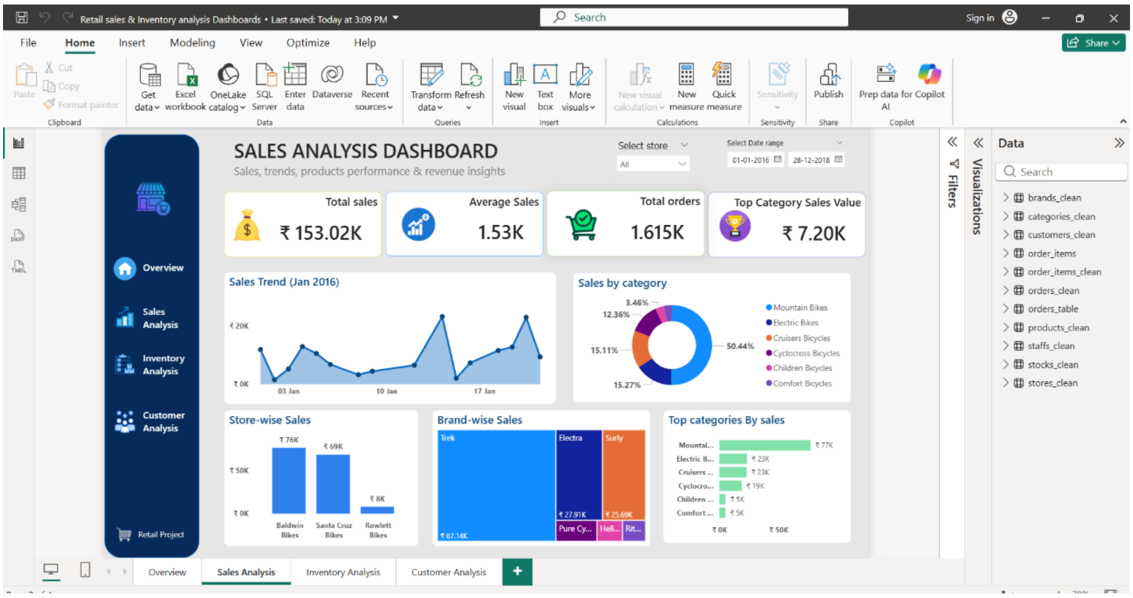
The dashboard showed that Mountain Bikes and Trek products generated the highest sales, while Baldwin Bikes was the best-performing store. It also highlighted inventory issues such as overstocking and low-stock products, helping the business understand where inventory management can be improved.

In addition, customer analysis revealed repeat purchasing behavior and high-performing customer regions, which can help the company improve customer retention and marketing strategies.

Overall, this project provides a practical business intelligence solution that helps management make faster, smarter, and more data-driven decisions for improving sales performance, inventory planning, customer satisfaction, and overall business growth.

THANK YOU


Retail sales & Inventory analysis Dashboards • Last saved: Today at 3:09 PM
RETAIL SALES DASHBOARD
Overview of Sales, Orders, Customers & Inventory
Select store
Select Date range
01-01-2016
28-12-2018
Total Customers
1.445K
Total orders
1.615K
Total products
321
Total sales
₹ 153.02K
Total Inventory
14K
Orders by stores
Sales by category
Top 5 products By Sales
Daily sales Trend
Inventory By Category
Overview
Sales Analysis
Inventory Analysis
Customer Analysis
Retail Project
Data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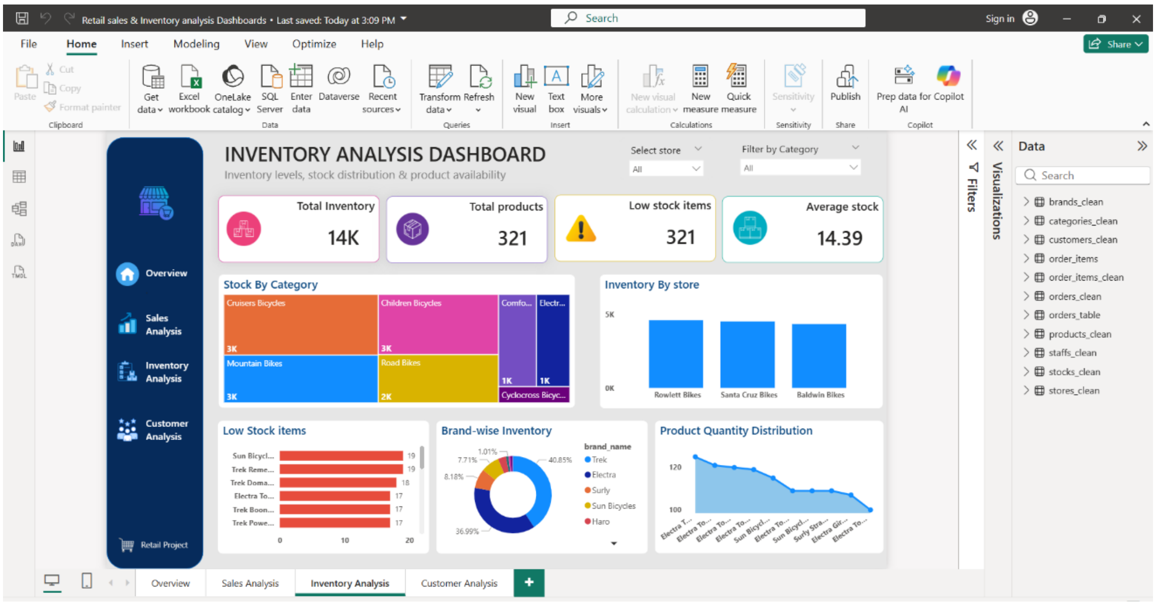
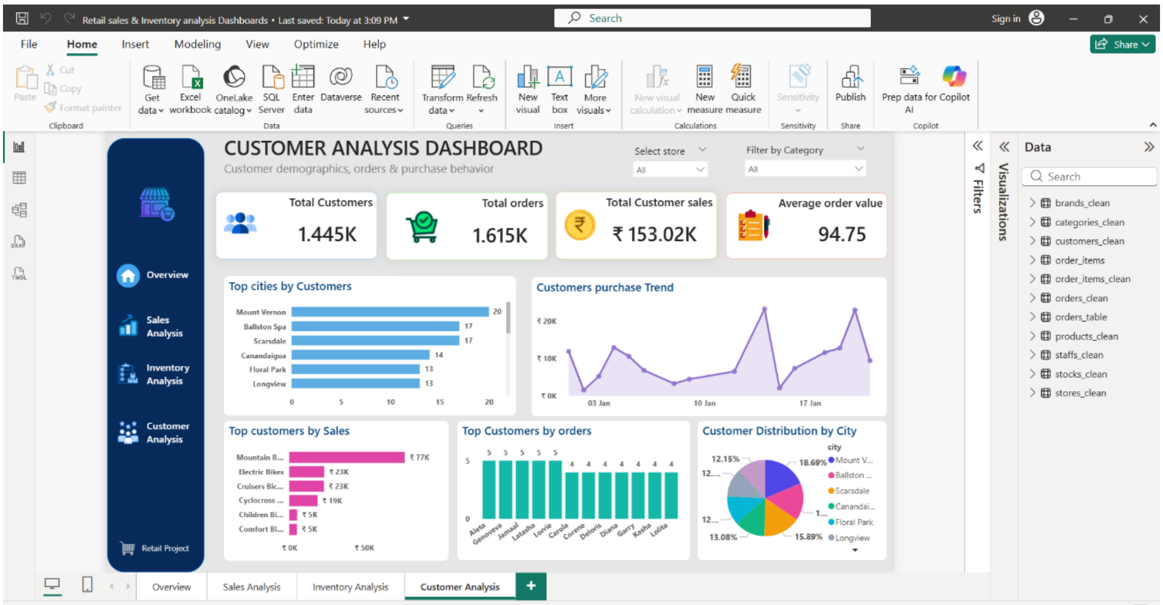
SALES ANALYSIS DASHBOARD
Sales, trends, products performance & revenue insights
Select store
Select Date range
01-01-2016
28-12-2018
Total sales
₹ 153.02K
Average Sales
1.53K
Total orders
1.615K
Top Category Sales Value
₹ 7.20K
Sales Trend (Jan 2016)
Sales by category
Store-wise Sales
Brand-wise Sales
Top categories By sales
Overview
Sales Analysis
Inventory Analysis
Customer Analysis
Retail Project
Data



INVENTORY ANALYSIS DASHBOARD
Inventory levels, stock distribution & product availability
Total Inventory
14K
Total products
321
Low stock items
321
Average stock
14.39
Stock By Category
Inventory By store
Low Stock items
Brand-wise Inventory
Product Quantity Distribution
Overview
Sales Analysis
Inventory Analysis
Customer Analysis
Retail Project



CUSTOMER ANALYSIS DASHBOARD
Customer demographics, orders & purchase behavior
Total Customers
1.445K
Total orders
1.615K
Total Customer sales
₹ 153.02K
Average order value
94.75
Top cities by Customers
Customers purchase Trend
Top customers by Sales
Top Customers by orders
Customer Distribution by City
Overview
Sales Analysis
Inventory Analysis
Customer Analysis
Retail Project